

Beta-Decay Effects on r-Process Nucleosynthesis in Neutron Star Mergers

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Motivation

The rapid neutron-capture process (r-process) is responsible for creating nearly half of the heavy elements found in the universe. The precise role played by nuclear physics inputs—such as β -decay rates—in shaping the final abundance pattern still remains uncertain. This study aims to investigate the impact of modified β -decay rates on r-process nucleosynthesis during a neutron star merger (NSM) event.

Theory

- The r-process (rapid neutron capture) occurs in high neutron density and temperature.
- Nuclei rapidly absorb neutrons and undergo β -decay to stabilize.
- β -decay rates control how quickly isotopes move toward stability — directly affecting element formation.
- Simulations rely on theoretical β -decay models due to lack of experimental data for neutron-rich nuclei.

Methods and Data

- Used PRISM to simulate 30 tracer particles from a high magnetic field neutron star merger (NES model).
 - Ran two sets of simulations:
 - One with standard Möller et al. (2003) β -decay rates
 - One with modified β -decay rates (reduced half-lives in neutron-rich regions)
 - Analyzed final abundances and β -decay rates
 - Compared results by averaging over all tracers and plotting differences in $Y(A)$, decay rates, and isotope distributions.

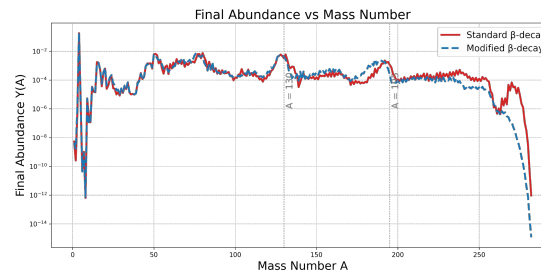


Figure 1: Final abundances ($Y(A)$) vs mass number A , averaged over 30 tracers. Solid line shows standard β -decay; dashed line shows modified rates. Key features such as the rare-earth peak and third peak are shifted under modified decay rates.

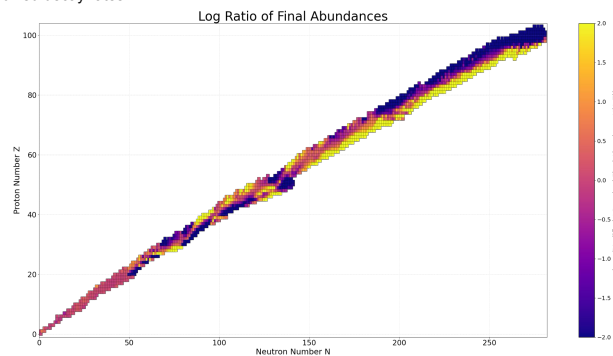


Figure 2: Logarithmic ratio of final abundances between modified and standard datasets from a high magnetic field neutron star merger simulation (NES model). Yellow regions reflect significantly more production under modified β -decay rates based on the methodology of Lund et al. (2023).

Results

- Modified β -decay rates produced notable differences in the heavy-mass region ($A > 200$).
- The final abundance peak near $A \approx 195$ is broadened and shifted.
- Logarithmic comparison of β -decay rates (Z, N) shows a systematic speed-up in decay times across neutron-rich nuclei.
- Modified rates increase the speed of matter flow back to stability, shortening the freeze-out period.

Conclusion

- β -decay physics has a measurable impact on r-process abundance patterns.
- This effect is most pronounced in heavy nuclei beyond the second r-process peak.
- Future models should explore broader nuclear datasets and uncertainty quantification.

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